

BIS KNOWLEDGE BASE: TEXTILES & LEATHER

Comprehensive AI/RAG Reference Manual for Department TXD (Bureau of Indian Standards)

Department Code: TXD | **Division Council:** Textiles Division Council (TXDC) | **Document Version:** 1.0 (Official Current Standards) | **Scope:** Natural/Synthetic Fibres, Geosynthetics, Medical & Protective Textiles, Leather & Footwear

DEPARTMENT AT A GLANCE (TXD):

- Published Standards:** Over 1,630 active Indian Standards (IS)
- Standards Under Development/Revision:** ~114 active projects
- Sectional Committees:** 25 Specialized Technical Committees (TXD 01 to TXD 37)
- Core Mandate:** Standardization in raw materials, testing methods, chemical processing, technical textiles, protective wear, eco-textiles, and leather/footwear items.

1. Department Scope & Responsibilities

The **Textiles and Leather Department (TXD)** operates under the governing authority of the **Textiles Division Council (TXDC)** at the Bureau of Indian Standards (BIS). TXD is responsible for formulating National Standards spanning the entire value chain of the textile and leather industries in India.

Primary Responsibilities & Key Scope Areas:

- Raw Materials & Fibres:** Standardization of natural fibres (cotton, wool, silk, jute, coir) and man-made/synthetic fibres (polyester, nylon, viscose, polypropylene, acrylic, aramid).
- Yarns, Fabrics & Garments:** Specifications for woven, knitted, non-woven, and braided fabrics, traditional garments, uniforms, handloom, and khadi products.
- Technical Textiles (12 Sub-domains):** Formulating crucial standards for *Agrotech, Meditech, Mobiltech, Oekotech, Packtech, Protech, Buildtech, Clothtech, Geotech, Hometech, Indutech, and Sportech*. Technical textiles are a primary thrust area under national regulatory Quality Control Orders (QCOs).
- Leather, Footwear & Leather Goods:** Raw hides, skins, finished leathers, leather chemicals, footwear components, industrial safety footwear, and consumer leather goods (TXD 18 / TXD 27).
- Testing Methodologies:** Standardizing physical, chemical, eco-testing, color fastness, optical, biological, and flammability testing methods for global harmonization.
- Machinery & Accessories:** Standardization of textile machinery, spinning/weaving accessories, dyehouse equipment, and testing equipment (TXD 09).
- Regulatory Support & Quality Control Orders (QCOs):** Assisting the Ministry of Textiles and Ministry of Commerce and Industry in implementing mandatory BIS certification (ISI Mark) under QCOs for geotextiles, protective textiles, medical textiles, and footwear.

2. Important Sectional / Technical Committees (TXD Committees)

Standardization within TXD is executed through specialized Sectional Committees comprising industry experts, academic institutions, research associations (e.g., SITRA, NITRA, BTRA, ATIRA, CLRI), and government authorities.

Committee Code	Committee Name	Technical Scope & Areas Covered
TXD 01	Physical Methods of Test	Standardization of physical test procedures for fibres, yarns, fabrics, nonwovens (tensile strength, tear resistance, air permeability, thickness, bursting strength).
TXD 03	Jute and Jute Products	Standards for raw jute, jute hessian, sacking bags, jute carpet backing, and eco-friendly jute packaging solutions.
TXD 04	Wool, Wool Products & Floor Coverings	Raw wool grading, woollen yarns, worsted fabrics, blankets, hand-knotted and machine-made textile floor coverings/carpets.
TXD 05	Chemical Methods of Test	Chemical analysis, blend composition determination, eco-testing (pH, formaldehydes, heavy metals, banned azo dyes), color fastness, and dimensional changes.
TXD 07	Textile Specialty Chemicals & Dyestuffs	Specifications and test methods for textile auxiliaries, eco-friendly dyes, finishing agents, sizing chemicals, and scouring agents.
TXD 08	Handloom and Khadi	Specifications for traditional handloom fabrics, khadi cotton/silk products, sarees, lungis, and artisan textile items.
TXD 09	Textile Machinery and Accessories	Machinery parts, spinning rings, shuttles, drop wires, heald frames, weaving looms, and finishing machines.
TXD 14	Silk, Silk Products and Coir	Raw silk grading, reeled silk yarns, silk fabrics, coir fibre, coir mats, coir mattresses, and rubberized coir products.
TXD 18	Leather, Footwear and Leather Products	Raw hides/skins, finished leathers, leather safety boots, dress shoes, sports shoes, footwear components, and leather travel goods.
TXD 30	Geotextiles and Industrial Fabrics	Geotextiles, geogrids, geomembranes, filtration fabrics, conveyor belting, filter fabrics, and civil engineering textiles.
TXD 31	Man-Made Fibres, Cotton and Products	Cotton fibre grading, polyester/viscose staple fibres, filament yarns, blended apparel fabrics, suiting, and shirting.
TXD 32	Protective Clothing / Textiles	Flame retardant clothing, firefighter protective gear, high-visibility garments, bullet resistant jackets, nuclear/chemical protective wear.
TXD 33	Industrial Textiles	Tyre cord fabrics, narrow fabrics, webbing, ropes, cordage, tarpaulins, tentage, and industrial sewing threads.
TXD 35	Agrotextiles	Shade nets, crop covers, mulch films, anti-hail nets, bird protection nets, and agricultural capillary mats.
TXD 36	Medical Textiles	Surgical masks (N95/medical), PPE coveralls, sanitary napkins, baby diapers, medical sponges, surgical sutures, and compression hosiery.
TXD 37	Sportect Textiles	High-performance sportswear, activewear, swimwear, mountaineering ropes, and sports equipment fabrics.

3. Major Products & Areas Covered

The TXD portfolio is classified into major product sectors:

- **Protective Textiles (Protech):** Firefighter gear, industrial heat-resistant suits, chemical protective clothing, high-visibility jackets, bullet-resistant body armor (IS 17051), ballistic helmets, tactical webbing.
- **Geotextiles & Geosynthetics (Geotech):** Woven and non-woven geotextiles for road reinforcement, slope stabilization, railway track sub-grade, erosion control, and drainage systems.
- **Medical & Hygiene Textiles (Meditech):** Surgical drapes/gowns, medical face masks, coveralls, absorbent cotton, sanitary napkins, adult diapers, compression garments.
- **Agrotextiles (Agrotech):** High-density shade nets for horticulture, insect nets, ground covers, crop covers, and tobacco harvester gloves.
- **Apparel & Uniforms:** Cotton/polyester blended fabrics, uniform suitings/shirtings, workwear, military camouflage uniforms, defense tactical pouches.
- **Leather & Footwear:** Industrial protective footwear with steel toe caps, lining leathers, upper leathers, rubber/polyurethane soles, sports footwear.

4. Key Indian Standards (IS Numbers & Titles)

Below is a curated repository of foundational Indian Standards formulated by TXD, essential for RAG query matching.

IS Code & Year	Exact Title of Standard	Committee	Certification
IS 17051:2018	Textiles — Body Armour — Specification	TXD 32	MANDATORY
IS 15748:2022	Textiles — Requirements for Clothing Made of Limited Flame Spread Materials and Material Assemblies Affording Protection Against Heat and Flame	TXD 32	MANDATORY
IS 16890:2018	Textiles — Protective Clothing for Firefighters — Specification	TXD 32	MANDATORY
IS 15809:2017	High Visibility Warning Clothes — Specification (First Revision)	TXD 32	MANDATORY
IS 15741:2007	Textiles — Resistance to Ignition of Curtains and Drapes — Specification	TXD 32	MANDATORY
IS 15768:2008	Textiles — Resistance to Ignition of Upholstered Composites Used for Non-Domestic Furniture — Specification	TXD 32	MANDATORY
IS 16655:2017	Textiles — Protective Clothing for Use in Welding and Allied Processes	TXD 32	MANDATORY
IS 16289:2014	Medical Textiles — Surgical Face Masks — Specification	TXD 36	MANDATORY
IS 17356:2020	Medical Textiles — Coveralls for COVID-19 / Pandemic Requirements — Specification	TXD 36	MANDATORY
IS 5405:2019	Sanitary Napkins — Specification (Second Revision)	TXD 36	MANDATORY
IS 16352:2020	Geotextiles — Polypropylene Woven Geotextiles for Sub-grade Stabilization in Highways — Specification	TXD 30	MANDATORY

IS Code & Year	Exact Title of Standard	Committee	Certification
IS 16391:2015	Geotextiles — Non-woven Geotextiles Used in Sub-grade Separation and Filtration — Specification	TXD 30	MANDATORY
IS 15298 (Pt 2):2016	Personal Protective Equipment - Part 2: Test Methods and Requirements for Safety Footwear	TXD 18	MANDATORY
IS 1969 (Pt 1):2018	Textiles — Tensile Properties of Fabrics — Part 1: Determination of Maximum Force and Elongation at Maximum Force Using the Strip Method	TXD 01	VOLUNTARY
IS 1954:2024	Determination of Length and Width of Woven Fabrics — Method	TXD 01	VOLUNTARY
IS 6359:2023	Method for Conditioning of Textiles (First Revision)	TXD 01	VOLUNTARY
IS 16008 (Pt 1):2016	Agro-textiles — Shade Nets for Agricultural and Horticultural Purposes — Part 1: Made from Shedding Nets	TXD 35	MANDATORY
IS 15852:2009	Textiles — Polyester Blended Woven Shirting for Uniforms — Specification	TXD 31	VOLUNTARY
IS 15853:2009	Textiles — Polyester Blended Woven Suiting for Uniforms — Specification	TXD 31	VOLUNTARY

5. Important Testing Standards & Methods

Testing methodology forms the core of verification for TXD standards. Testing is divided into physical, chemical, and specialized protective performance evaluations.

A. Physical & Mechanical Testing (TXD 01)

- **IS 1969 (Part 1):2018** — Tensile strength and elongation test using Strip Method. Essential for apparel and industrial canvas.
- **IS 1969 (Part 2):2018** — Determination of breaking force using Grab Method.
- **IS 6489 (Part 1 & 2)** — Tear resistance of woven fabrics using Elmendorf falling pendulum apparatus and tongue tear test.
- **IS 1954:2024** — Methods for determination of length and width of woven fabrics.
- **IS 6359:2023** — Standard conditioning atmosphere for textile testing (typically 20 ± 2 °C and 65 ± 4 % Relative Humidity).

B. Chemical, Eco-Testing & Color Fastness (TXD 05)

- **IS 10260** series — Methods for determination of color fastness of textiles to washing, light, rubbing (crockmeter), perspiration, and dry cleaning.
- **IS 6330:2021** — Domestic washing and drying procedures for textile testing.
- **IS 1299:2026** — Determination of dimensional changes on washing of fabrics woven from rayon and synthetic fibres.
- **Eco-Testing Methods:** Determination of formaldehyde content, pH value of aqueous extract (**IS 1390**), and detection of banned aromatic amines (Azo dyes) using GC-MS/HPLC equipment.

C. Thermal & Flame Protection Tests (TXD 32)

- **IS 17468:2020** — Clothing and Equipment for Protection against Heat — Test Method for Convective Heat Resistance using a Hot Air Circulating Oven.
- **Limited Flame Spread Test:** Evaluates vertical flame propagation, flame-through, and afterglow duration when subjected to direct burner flame.
- **Thermal Protection Performance (TPP):** Measures combined convective and radiant heat transfer indices to calculate burn prevention threshold.

6. BIS Certification & Compliance Requirements

Certification under TXD follows the **Product Certification Scheme (ISI Mark Scheme - Scheme I, Schedule II)** governed by the Bureau of Indian Standards Act, 2016.

IMPORTANT REGULATORY MANDATE (QCOs): The Ministry of Textiles and Ministry of Commerce have issued Quality Control Orders (QCOs) making BIS Certification compulsory for specific sub-sectors. Uncertified manufacture, import, or sale of these items in India is illegal under BIS Law.

A. Key Mandatory QCO Schemes in Textiles & Leather

1. **Technical Textiles (Protective & Agrotextiles QCOs):** Covers 12 protective textile products (IS 17051, IS 15748, IS 16890, IS 15809, IS 15741, etc.) and 6 agrotextile products.
2. **Geotextiles QCO:** Mandates ISI mark for geotextiles used in roads, railways, erosion control, and slopes (IS 16352, IS 16391).
3. **Medical Textiles QCO:** Mandates certification for surgical face masks (IS 16289), sanitary napkins (IS 5405), baby diapers, and protective coveralls.
4. **Footwear Products QCO:** Mandates ISI marking under **IS 15298 (Parts 2, 3, 4)** for safety, protective, and occupational footwear.

B. Compliance & Audit Workflow for Manufacturers

- **In-house Testing Infrastructure:** Manufacturers must set up complete testing laboratory facilities as specified in the BIS *Scheme of Inspection and Testing (SIT)* for each standard.
- **Factory Inspection:** BIS Officers perform physical audits of manufacturing facilities, process control, raw material quality, and calibration of test equipment.
- **Sample Verification:** Dual testing — sample drawn during factory audit is tested in-house and simultaneously sent to an independent BIS-Recognized Third-Party Laboratory.
- **License Validity & Grant:** Licenses are issued initially for 1 to 2 years and are renewable subject to satisfactory surveillance audits and market sample testing.
- **Foreign Manufacturers Certification Scheme (FMCS):** Overseas manufacturers supplying to India must obtain a BIS FMCS license by appointing an Authorized Indian Representative (AIR).

7. Tracked Revisions, Superseded & Withdrawn Standards

To ensure RAG accuracy and prevent hallucination, the following table tracks updated and superseded versions of key standards under TXD:

Current Active Standard	Superseded / Revised Standard	Key Technical Revisions / Changes
IS 15748:2022	IS 15748:2007	Aligned with international ISO 11612 standards; expanded thermal protection index testing, flame spread criteria, and seam strength requirements.
IS 15809:2017	IS 15809:2008	Updated minimum retroreflective tape surface area, color chromaticity coordinates after washing, and ergonomic requirements.
IS 1299:2026	IS 1299:1984	Third revision; modernized washing procedures and automated measuring equipment requirements for synthetic woven fabrics.
IS 2977:2026	IS 2977:1989	Updated water soaking dimensional stability measurement protocols for non-woollen fabrics.
IS 665:2026	IS 665:1989	Updated procedure for measuring dimensional relaxation of woollen fabrics upon water immersion.
IS 5405:2019	IS 5405:1980 / 2003	Second revision; added bio-degradability guidelines, absorbency testing updates, micro-biological safety limits, and disposal criteria.

8. Product-to-Standard Mapping Matrix

Quick lookup matrix for mapping real-world products to exact Indian Standard specifications and responsible technical committees.

Product Category / Name	Applicable Indian Standard	Committee	Core Testing Requirements
Bulletproof Jackets / Body Armour	IS 17051:2018	TXD 32	Ballistic resistance (small arms/rifle), trauma depth, fragmentation resistance, weight limits.
Firefighter Protective Suits	IS 16890:2018	TXD 32	Heat transfer (convective & radiant), flame propagation, water resistance, seam strength.
Industrial Heat/Flame Apparel	IS 15748:2022	TXD 32	Limited flame spread, contact heat resistance, molten metal splash resistance.
High-Visibility Safety Vests	IS 15809:2017	TXD 32	Luminous factor, chromaticity coordinates, retroreflective performance after 25 wash cycles.
Highway Sub-grade Geotextiles	IS 16352:2020	TXD 30	Tensile strength, puncture resistance, CBR burst strength, apparent opening size (AOS), water permeability.
Surgical Masks (3-Ply / Medical)	IS 16289:2014	TXD 36	Bacterial Filtration Efficiency (BFE), differential pressure (breathability), splash resistance.

Product Category / Name	Applicable Indian Standard	Committee	Core Testing Requirements
Industrial Safety Boots	IS 15298 (Pt 2):2016	TXD 18	Impact resistance (200 Joules steel toe cap), compression resistance, sole slip resistance, upper tear strength.
Horticulture Shade Nets	IS 16008 (Pt 1):2016	TXD 35	Shade factor percentage, UV weathering endurance, breaking force, burst strength.
Uniform Shirting (Poly-Cotton)	IS 15852:2009	TXD 31	Blend ratio accuracy, color fastness to washing/light, dimensional stability, pilling resistance.

9. Common Questions & Answers (AI RAG Embeddings)

Q1: Which Indian Standard governs body armour and bulletproof jackets in India, and what are its key requirements?

Answer: Body armour and bulletproof jackets are governed by **IS 17051:2018** (*Textiles — Body Armour — Specification*), formulated by sectional committee **TXD 32**. It defines protection levels against small arms and high-velocity ammunition, specifying strict test methods for ballistic impact resistance, maximum allowable blunt trauma backface signature depth, ergonomics, and environmental conditioning performance. BIS certification under IS 17051 is mandatory for defense and law enforcement procurement in India.

Q2: Is BIS certification compulsory for protective textile garments used in high-heat industrial environments?

Answer: Yes. Under the Quality Control Order (QCO) issued for Technical Textiles (Protective Textiles), protective clothing for industrial workers exposed to heat must strictly conform to **IS 15748:2022** (*Textiles — Requirements for Clothing Made of Limited Flame Spread Materials and Material Assemblies Affording Protection Against Heat and Flame*) and bear the ISI Mark. Non-compliant garments cannot be legally manufactured, imported, or sold in India.

Q3: What standard applies to surgical face masks in India, and what are the mandatory test parameters?

Answer: Surgical face masks are covered under **IS 16289:2014** (*Medical Textiles — Surgical Face Masks — Specification*) developed by **TXD 36** (Medical Textiles). Essential testing parameters include Bacterial Filtration Efficiency (BFE $\geq 95\%$ or $\geq 98\%$ depending on Class), Differential Pressure / ΔP (measuring breathability), Particle Filtration Efficiency, and Microbial Cleanliness.

Q4: What is the standard test method for determining fabric tensile strength under BIS TXD?

Answer: The primary physical testing standard for fabric tensile properties is **IS 1969 (Part 1):2018** (*Textiles — Tensile Properties of Fabrics — Part 1: Determination of Maximum Force and Elongation at Maximum Force Using the Strip Method*), formulated by committee **TXD 01**. It replaces older revisions and aligns with international strip method test protocols.

Q5: How does BIS classify safety footwear under the Textiles and Leather department?

Answer: Safety footwear is standardized by Sectional Committee **TXD 18** (Leather, Footwear and Leather Products) under **IS 15298 (Part 2):2016** (*Personal Protective Equipment - Part 2: Test Methods and Requirements for Safety Footwear*). It mandates features like impact resistance of toe caps (200 Joules), penetration resistance, oil/slip resistance, and electrical insulation or antistatic properties.

10. Search Keywords & Synonyms for Vector RAG Ingestion

To maximize semantic retrieval, dense vector indexing, and cross-keyword matching in RAG pipelines, use the following structured metadata tag sets:

Primary Department Identifiers:

BIS TXD

Textiles Division Council

TXDC

Bureau of Indian Standards Textile Department

TXD 01

TXD 18

TXD 30

TXD 32

TXD 36

Technical Textiles Terms:

Protech

Geotech

Meditech

Agrotech

Buildtech

Indutech

Geotextiles

Polypropylene geotextile

Flame retardant clothing

Bulletproof jacket standard

IS 17051

IS 15748

IS 16890

IS 16289

IS 15298

Testing & Compliance Keywords:

Strip Method Tensile Test

IS 1969

Quality Control Order QCO

ISI Mark Scheme I

Scheme of Inspection and Testing SIT

BFE Bacterial Filtration Efficiency

Thermal Protection Performance TPP

Azo dye testing IS 1390

Conditioning atmosphere IS 6359

Product Synonyms & Cross-References:

Body armour

Firefighter suit

High vis vest

Safety boots steel toe

Surgical face mask

Sanitary pad standard IS 5405

Shade net IS 16008

Uniform shirting IS 15852

FMCS foreign manufacturer certification